

YITIAN WANG

ywang1057@ucr.edu ◇ <http://cosmotim.github.io> ◇ <http://linkedin.com/in/tim-wang-yitian/>

EDUCATION

- University of California, Riverside, CA** *Mar. 2021 - Jun. 2026 (expected)*
Ph.D. in Electrical and Computer Engineering
Award: MRS 2023 Spring Highly Commended Student Talk Award
Dissertation Completion Fellowship Award
- Columbia University in the City of New York, NY** *Aug. 2019 - Feb. 2021*
Master of Science, Material science and engineering
- University of California, Berkeley, CA** *Jul. 2017 - Aug. 2017*
Summer Session
- Beijing Normal University, CN** *Sep. 2015 - Jun. 2019*
Bachelor of Science, Physics
Award: 'Jingshi' Scholarship (2016 - 2018)

SKILLS

Software / Programming

- Python (advanced), MATLAB (advanced), SolidWorks (intermediate)
- GSAS-II (advanced), Mantid (advanced), ImageJ (intermediate)

Experimental Techniques

- FZ Crystal Growth (advanced), PVD (intermediate), CVD (intermediate)
- SEM (advanced), XRD (advanced), INS (advanced), Raman (intermediate), FTIR (intermediate)
- PPMS (advanced), DSC (intermediate), TGA (intermediate)
- Battery Assembly (advanced), Impedance Spectroscopy (intermediate)

PROJECTS

- Thermal transport and ionic mobility in solid electrolytes** *Jun. 2021 - Jun. 2025*
University of California *Riverside, CA*
- Grew single crystal samples with floating zone method using image furnace.
 - Studied the physical properties of various solid electrolytes.
- Inelastic neutron scattering on LLZTO single crystal** *Jan. 2022 - Jul. 2023*
Oak Ridge National Laboratory *Oak Ridge, TN*
- Conducted inelastic neutron scattering experiments using TAX and ARCS.
 - Processed and analyzed the data with Mantid and Python.
- Two-channel fitting model for thermal conductivity** *Mar. 2022 - Sep. 2023*
University of California *Riverside, CA*
- Developed an expandable fitting model for thermal conductivity data in MatLab.

Advanced separator of lithium-ion battery*Columbia University**Oct. 2019 - Jan. 2021**New York, NY*

- Made composite battery separator with $\gamma - C_3N_4$ and PVDF.
- Assembled full cell batteries and tested electrochemical properties.

First-principle calculation of olivine structure*Columbia university**Sep. 2019 - Jan. 2020**New York, NY*

- Calculated the stable olivine structure with hydrogen defects with Quantum ESPRESSO.

Low temperature uniaxial strain device*Beijing Normal University**Sep. 2018 - May 2019**Beijing, CN*

- Modeled devices in Solidworks to apply uniaxial pressure on film samples utilizing thermal expansion.
- Tested the strain induced detwinning effect with PPMS.

SELECTED PUBLICATIONS

- Glass-like thermal transport in polycrystalline perovskite lithium-ion conductor $Li_{3/8}Sr_{7/16}Hf_{1/4}Ta_{3/4}O_3$
 - Chemical Communications, 2025
 - **Y. Wang**, Q. Jia, S. Li, L. Shi, Y. Li, X. Chen
- Low Thermal Conductivity and Lattice Anharmonicity of NaSICON-type Solid Electrolyte $Na_3Zr_2Si_2PO_{12}$
 - Tungsten, 2025 (Accepted)
 - **Y. Wang**, Q. Jia, S. Li, L. Shi, Y. Li, X. Chen
- Origin of intrinsically low thermal conductivity in a garnet-type solid electrolyte: Linking lattice and ionic dynamics with thermal transport
 - PRX Energy, 2025
 - **Y. Wang**, Y. Su, J. Carrete, H. Zhang, N. Wu, Y. Li, H. Li, J. He, Y. Xu, S. Guo, Q. Cai, D. L. Abernathy, T. Williams, K. V. Kravchyk, M. V. Kovalenko, G. K. H. Madsen, C. Li, X. Chen
- Thermal properties and lattice anharmonicity of Li-ion conducting garnet solid electrolyte $Li_{6.5}La_3Zr_{1.5}Ta_{0.5}O_{12}$
 - Journal of Materials Chemistry A, 2024
 - **Y. Wang**, S. Li, N. Wu, Q. Jia, T. Hoke, L. Shi, Y. Li, X. Chen
- Enhanced magnon thermal transport in yttrium-doped spin ladder compounds $Sr_{14-x}Y_xCu_{24}O_{41}$
 - Journal of Applied Physics, 2024
 - S. Li, S. Guo, **Y. Wang**, H. Li, Y. Xu, V. Carta, J. Zhou, X. Chen
- Size-dependent magnon thermal transport in a nanostructured quantum magnet
 - Cell Reports Physical Science, 2024
 - S. Guo, H. Li, X. Bai, **Y. Wang**, S. Li, R. E. Dunin-Borkowski, J. Zhou, X. Chen
- Crystal structure and thermoelectric properties of layered van der Waals semimetal $ZrTiSe_4$
 - Chemistry of Materials, 2022

- Y. Xu, Z. Barani, P. Xiao, S. Sudhindra, **Y. Wang**, A. A. Rezaie, V. Carta, K. N. Bozhilov, D. Luong, B. P. T. Fokwa, F. Kargar, A. A. Balandin, X. Chen
- Single crystal growth and electrochemical studies of garnet-type fast Li-ion conductors
 - Tungsten, 2022
 - **Y. Wang**, X. Chen

EXTRA-CIRRICULAR

GSA Python Coding Contest	Aug. 2020
COMAP Mathematical Contest in Modeling	Jan. 2017
BNU Physics Department Soccer Team	2015 - 2019